

SEC 1_FA1-GROUP 6-TAN,H

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```
results<-read.csv("results.csv")
skew_formula<-function(x){
mean_x<-mean(x,na.rm=TRUE)
sum2<-sum((x-mean_x)^2,na.rm=TRUE)
sum3<-sum((x-mean_x)^3,na.rm=TRUE)
skew<-(sqrt(sum(!is.na(x)))*sum3)/(sum2^(1.5))
return(skew)
}
c(Arch1=skew_formula(results$arch1),Prog1=skew_formula(results$prog1),
Arch2= skew_formula(results$arch2),Prog2=skew_formula(results$prog2))
```

```
##      Arch1      Prog1      Arch2      Prog2
## -0.5070824 -0.3443763  0.4623552 -0.2807343
```

The four examination subjects show different patterns of skewness. Based on the calculated skewness coefficients, Arch1, Prog1, and Prog2 are negatively skewed, meaning their distributions have a longer tail toward the lower scores. Meanwhile, Arch2 is positively skewed, meaning its distribution has a longer tail toward the higher scores. Since all the skewness coefficients are relatively close to zero, the distributions are only slightly to moderately skewed and are not extremely asymmetric.

```
pearson_skew<-function(x){
x<-x[!is.na(x)]
mean_x<-mean(x)
median_x<-median(x)
sd_x<-sd(x)
skew<-3 *(mean_x-median_x)/sd_x
return(skew)
}
c(Arch1=pearson_skew(results$arch1),Prog1=pearson_skew(results$prog1),
Arch2=pearson_skew(results$arch2), Prog2=pearson_skew(results$prog2))
```

```
##      Arch1      Prog1      Arch2      Prog2
## -0.5896811 -0.6345275  0.5190494 -0.3854051
```

Although there are some differences on the values, Pearson's formula gives a reasonable approximation of the skewness coefficient because its result is generally close to the values obtained from the skewness program; they both show the same direction of skewness.

```
female<-c(57,59,78,79,60,65,68,71,75,48,51,55,56,41,43,
44,75,78,80,81,83,83,85)
male<-c(48,49,49,30,30,31,32,35,37,41,86,42,51,53,56,
42,44,50,51,65,67,51,56,58,64,64,75)
stem(female)

##
##  The decimal point is 1 digit(s) to the right of the |
##
##  4 | 1348
##  5 | 15679
##  6 | 058
##  7 | 155889
##  8 | 01335

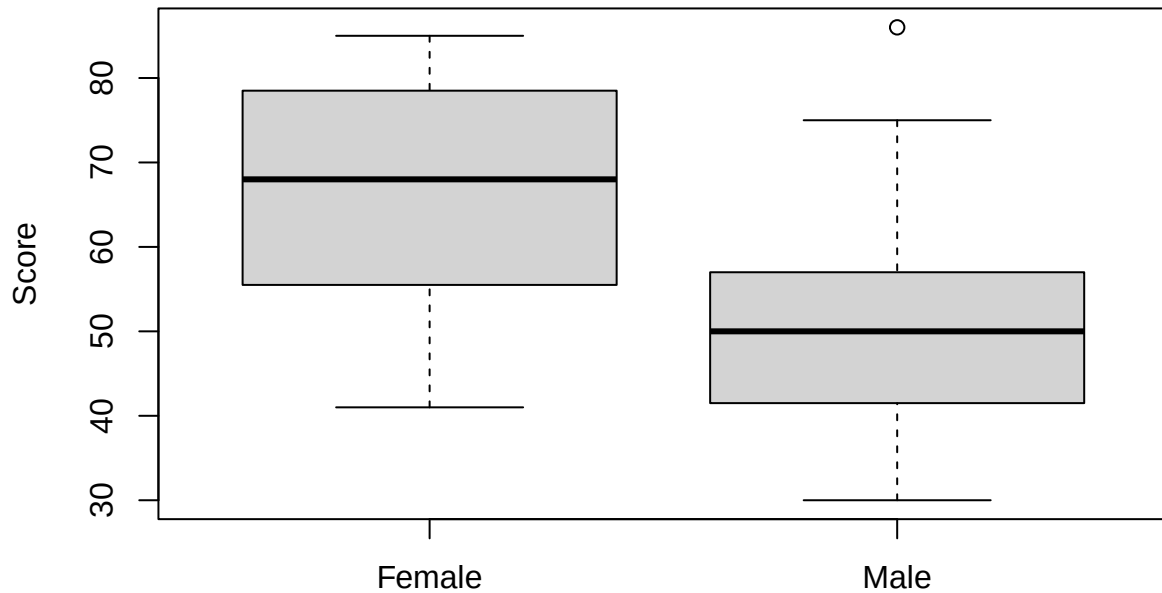
stem(male)
```

```
##
##  The decimal point is 1 digit(s) to the right of the |
##
##  3 | 001257
##  4 | 1224899
##  5 | 01113668
##  6 | 4457
##  7 | 5
##  8 | 6
```

A stem-and-leaf display shows the distribution of the data while still retaining the individual values of the observations. It allows the original data to be seen, unlike a histogram, where individual scores are grouped into intervals and their exact values cannot be identified.

```
boxplot(female, male, names = c("Female", "Male"),
        main = "Java Examination Scores", ylab = "Score")
```

Java Examination Scores



The box plots show that the female students performed better than male students in the Java programming examination. The median score of the females is higher than those of the males. The female box is also positioned higher on the graph, showing that most of their scores are higher than those of the males. On the other hand, the male scores are more spread across the lower part of the graph, with their scores ranging from 30 to 86, while the female scores range from 41 to 85.